

Customization + extensibility

Part of "Introduction to Positron" workshop



Posit Software, PBC



Settings

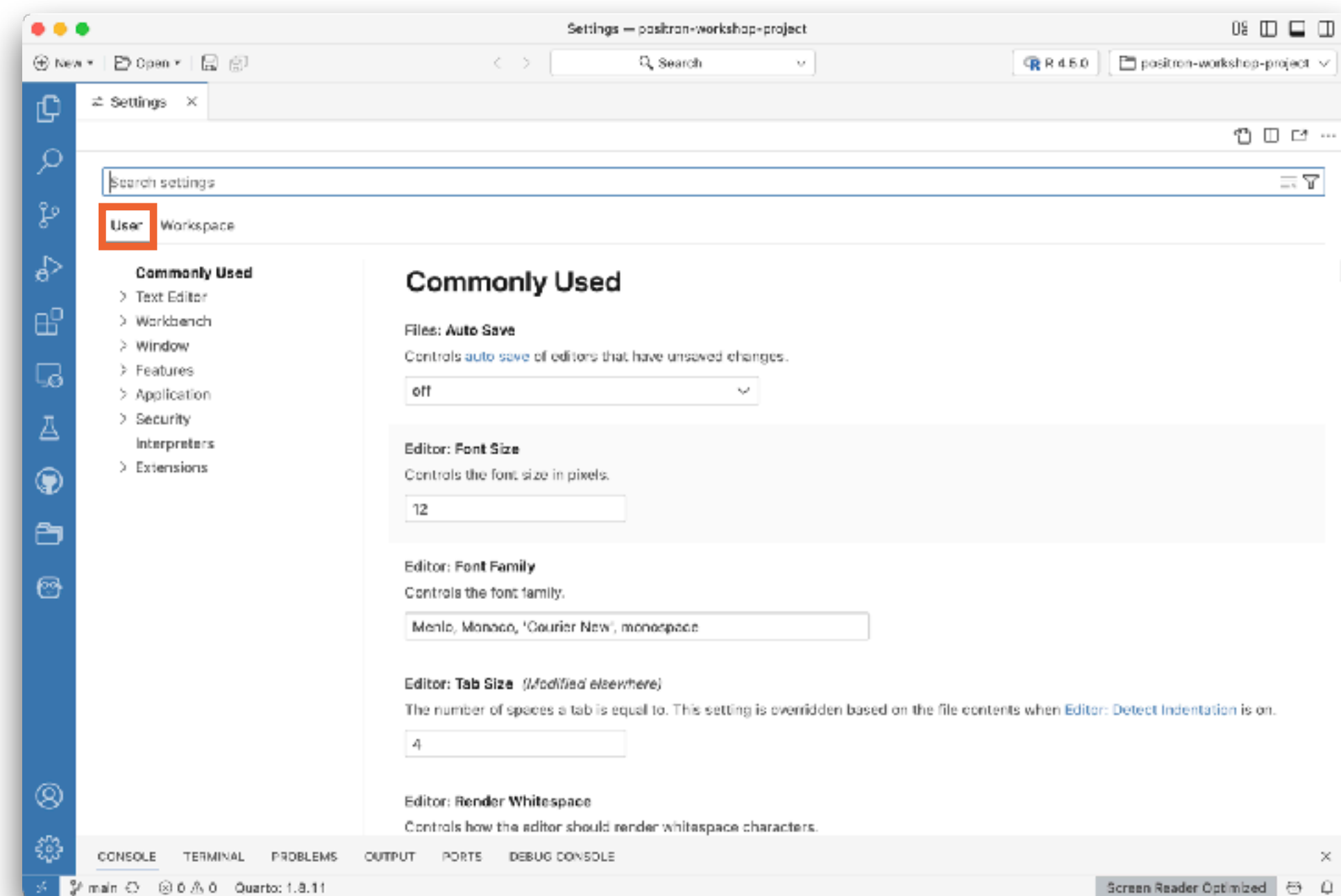
Settings

- Nearly every part of Positron's editor, user interface, and functional behavior has customizable options.
- Settings can be customized at two different scopes:
 - **User settings:** Settings that apply globally to any instance of Positron you open.
 - **Workspace settings:** Settings stored inside your workspace (folder) and only apply when the workspace is opened.
- Positron stores setting values in a settings JSON file.
- Settings can be changed by:
 - editing the settings JSON file or
 - using the Settings editor interface

Updating user settings

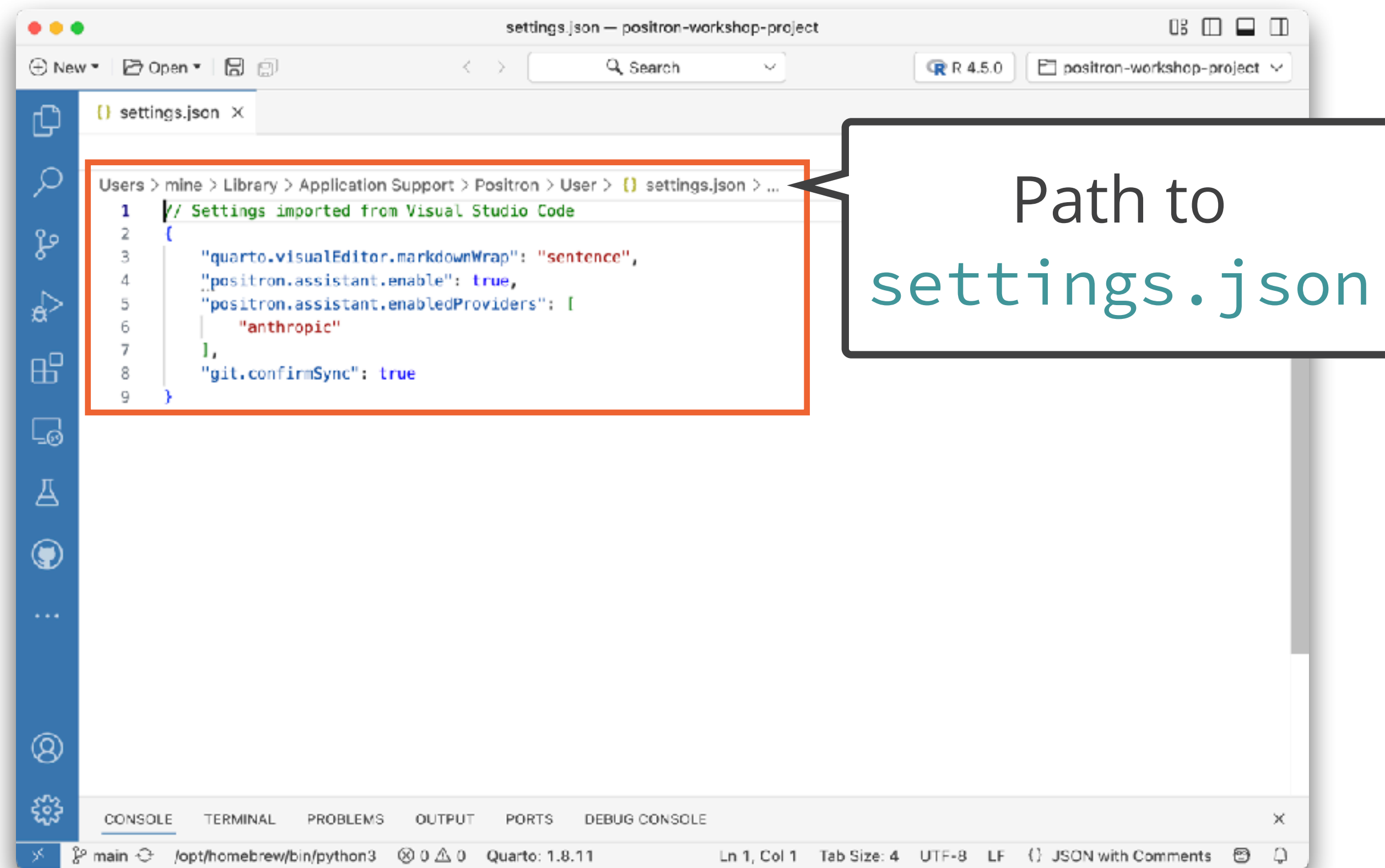
To update using the UI:

- In the Command Palette:
Preferences: Open User Settings
- In the top menu:
Positron > Settings... > Settings



To edit the JSON file:

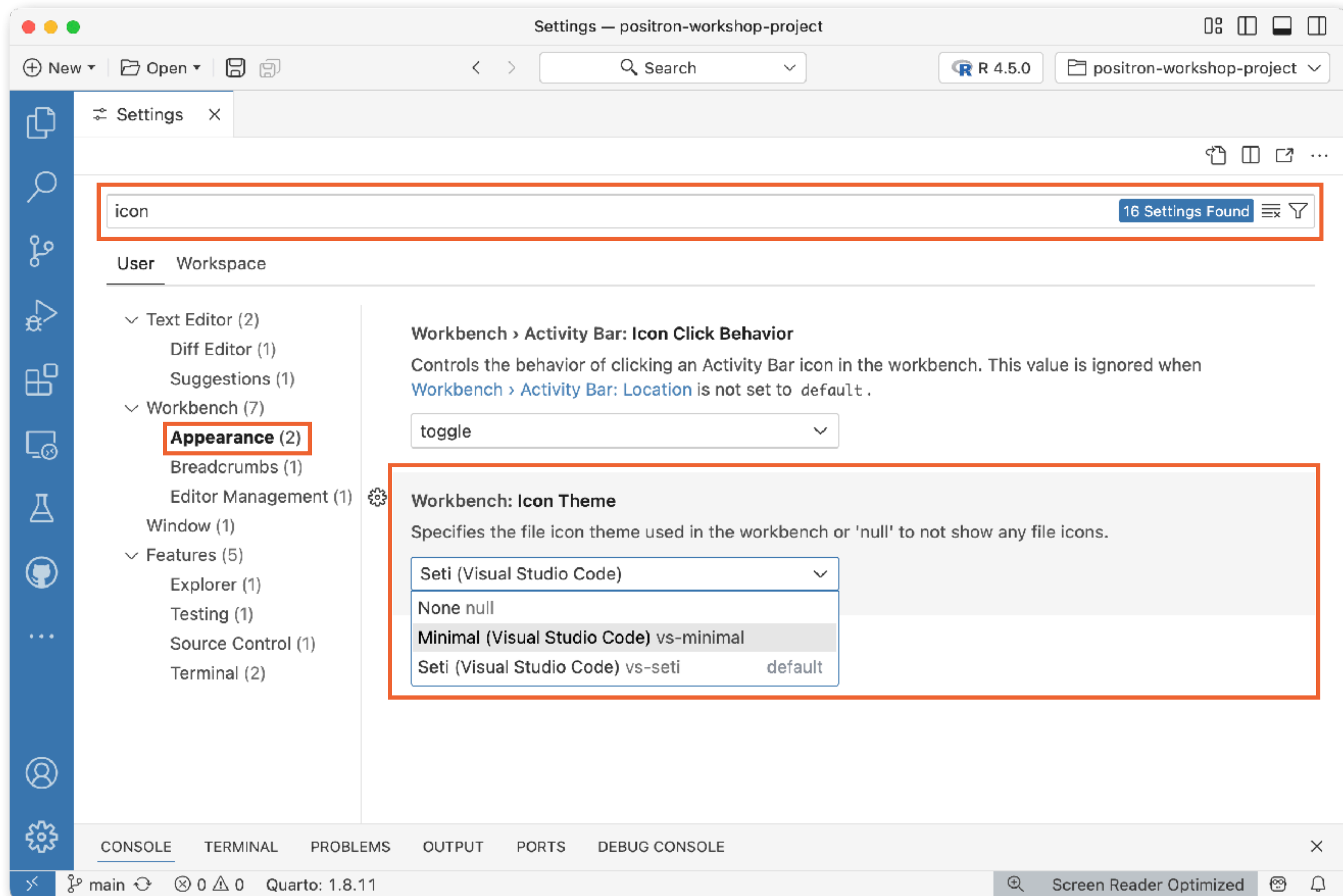
In the Command Palette:
Preferences: Open User Settings (JSON)

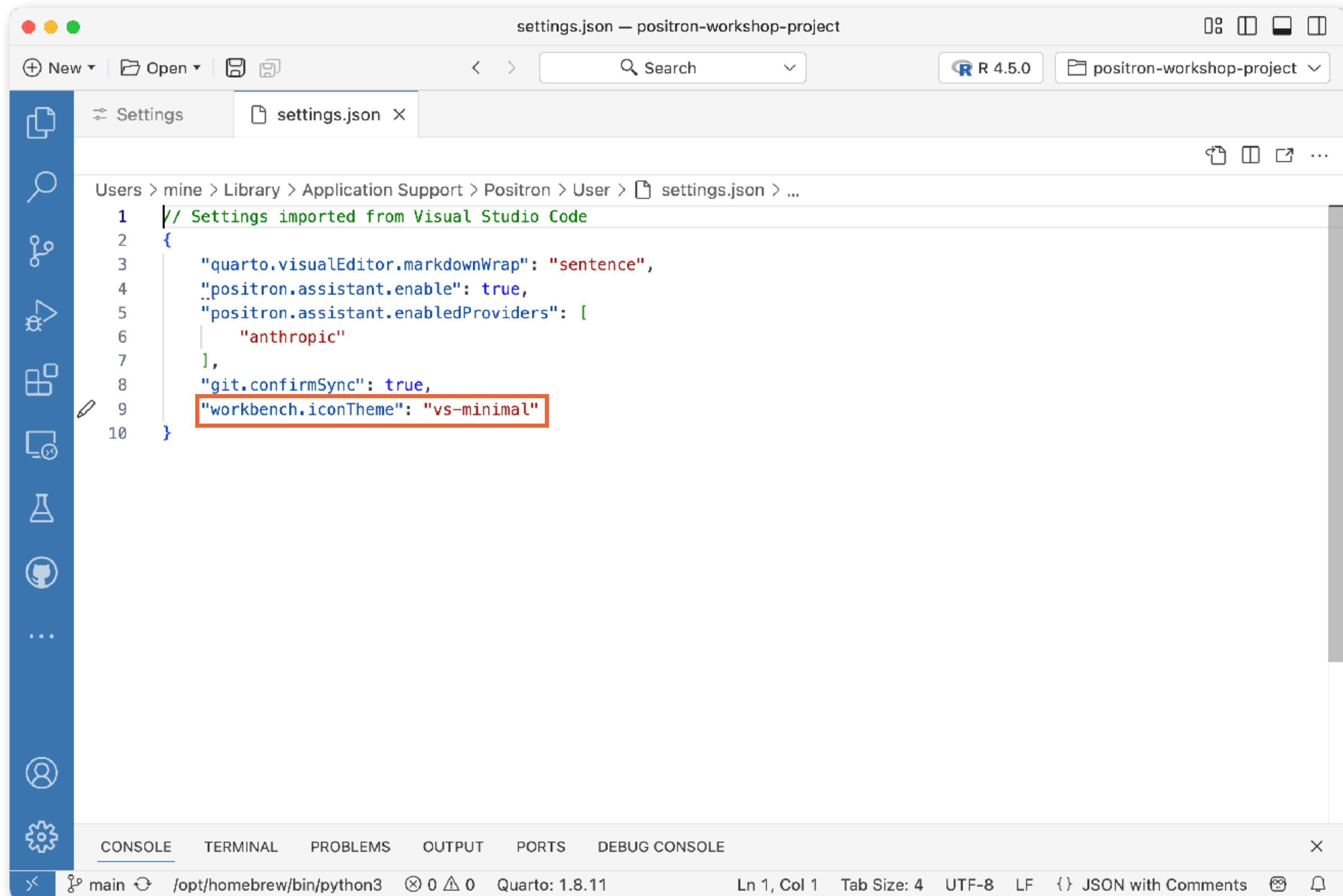


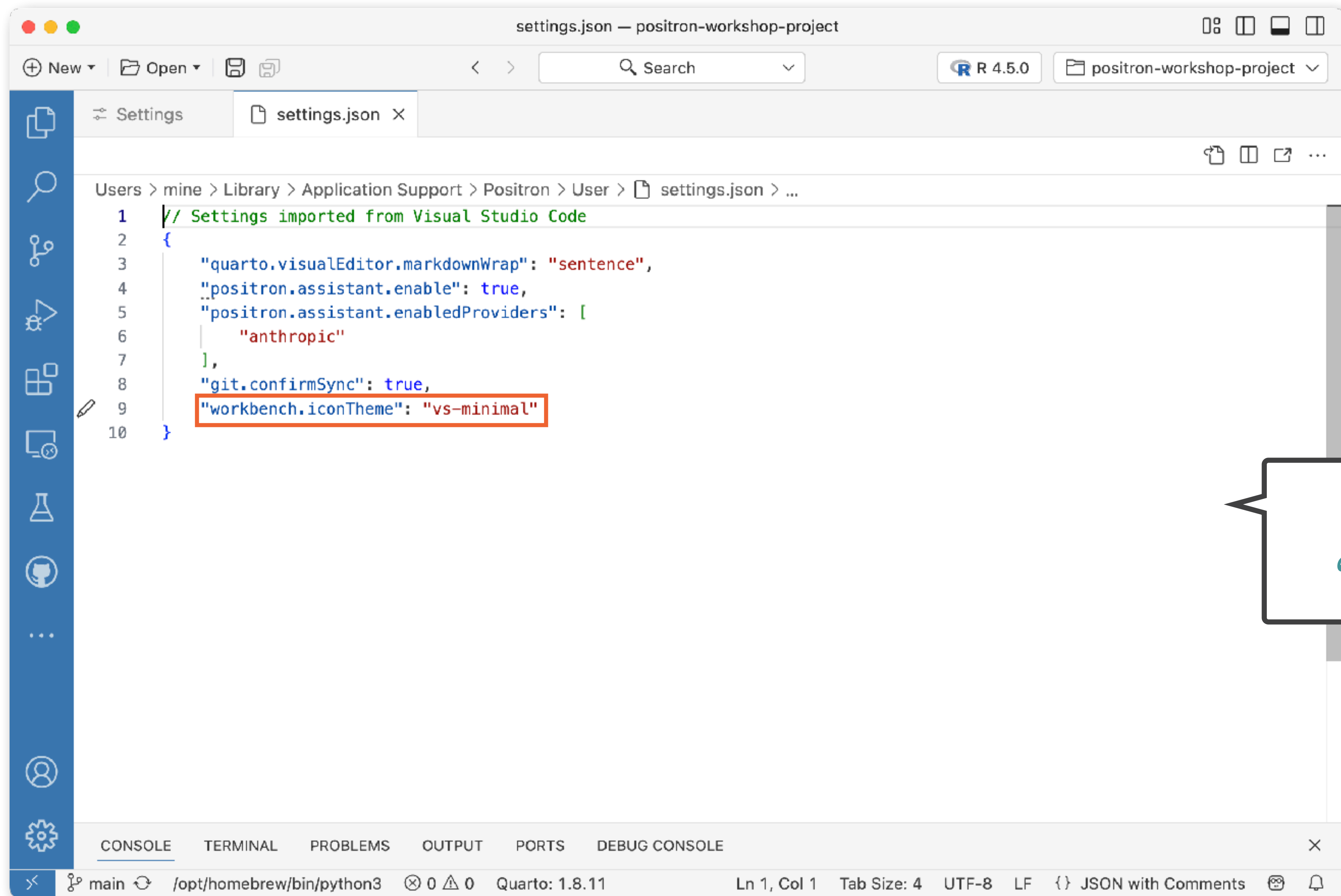
Your turn

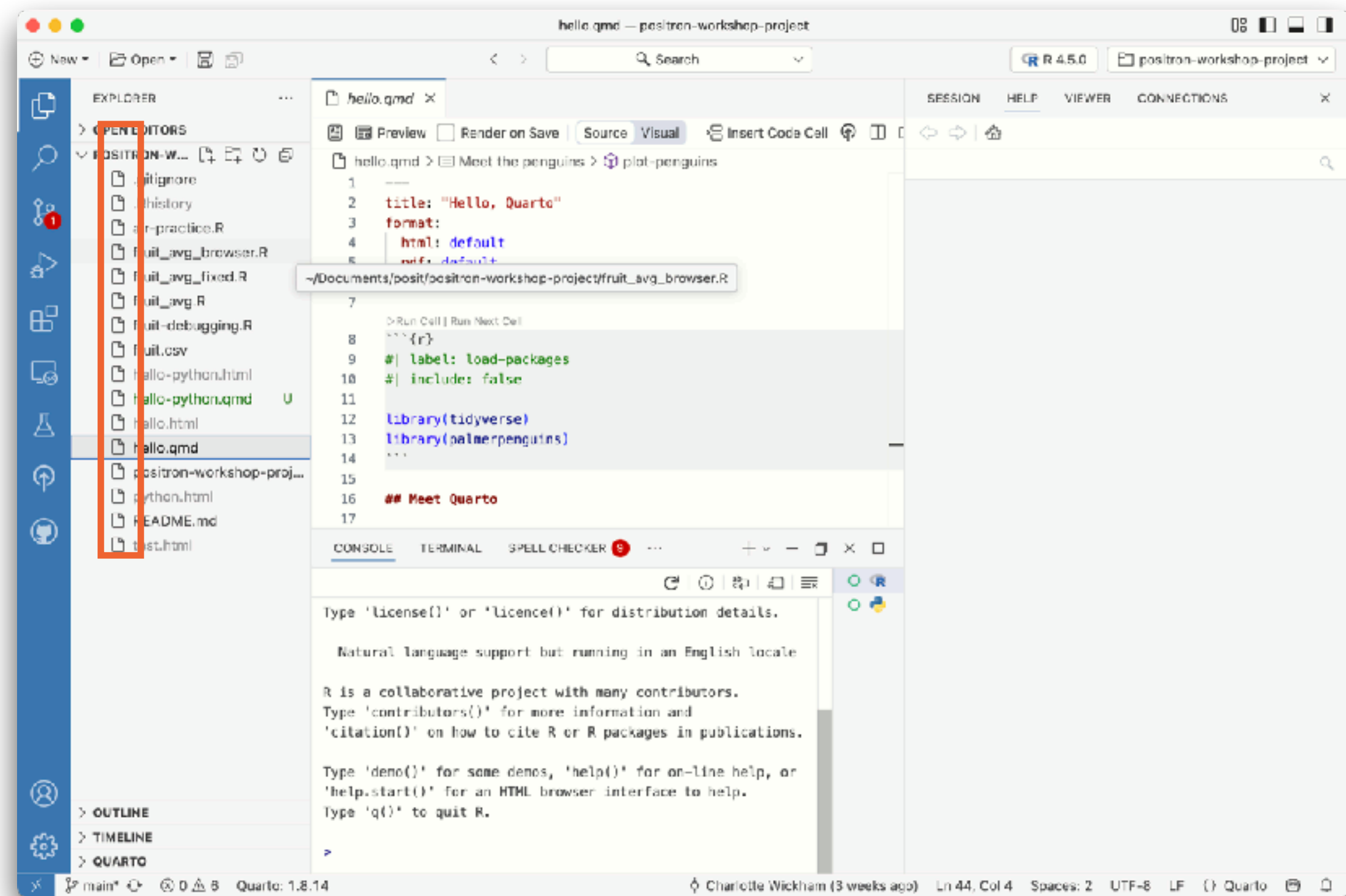
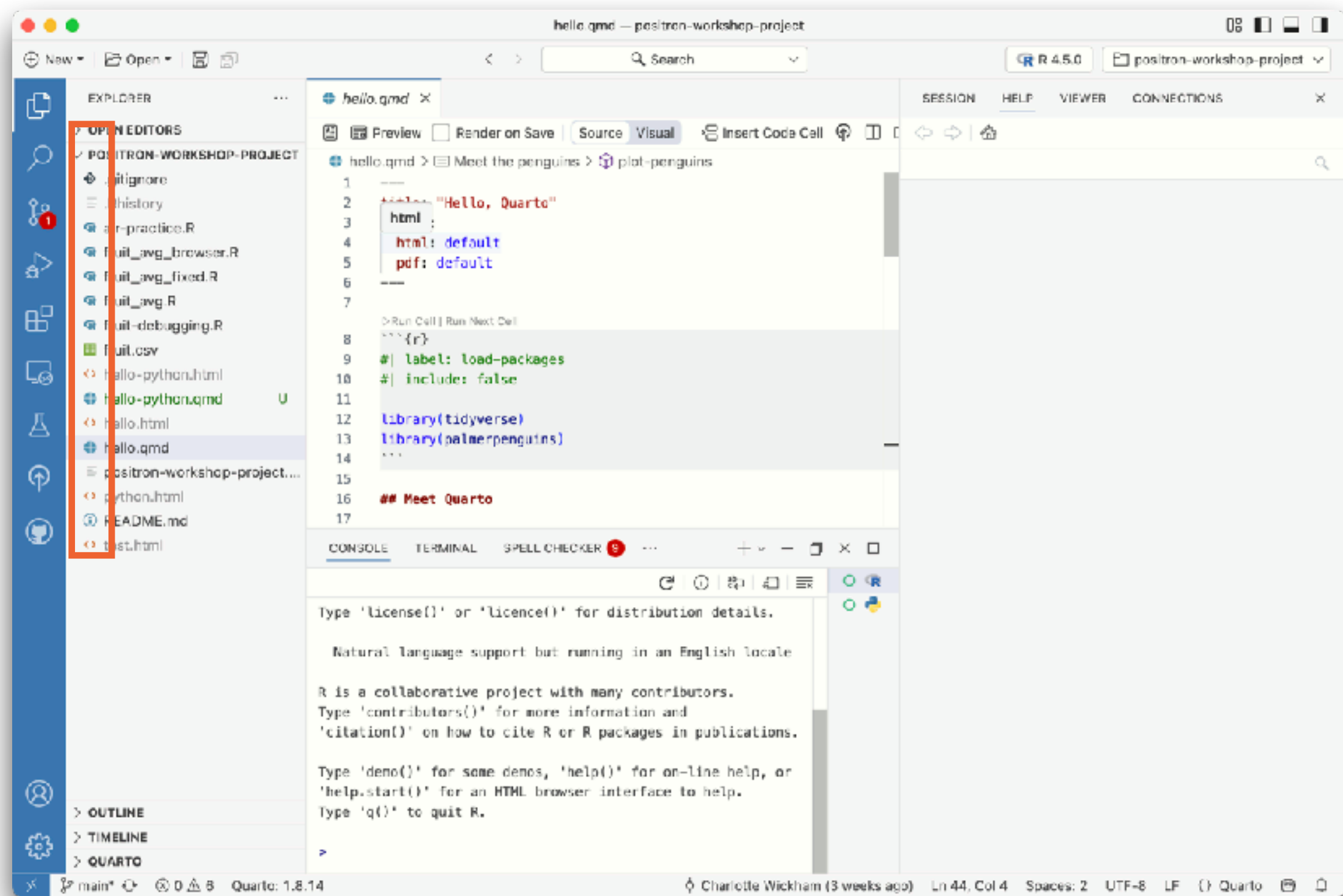
Customize user settings

- Update your user settings to change the Icon Theme to Minimal via the Settings editor interface.
 - Hint: Filter for settings with the word “icon”.
- Open the `user settings.json` file to confirm the change you made.
- In the `user settings.json` file, change the Icon Theme to `vs-seti` and save your changes.
- Go back to the Icon Theme settings in the Settings editor interface and confirm the change you made in the JSON file.









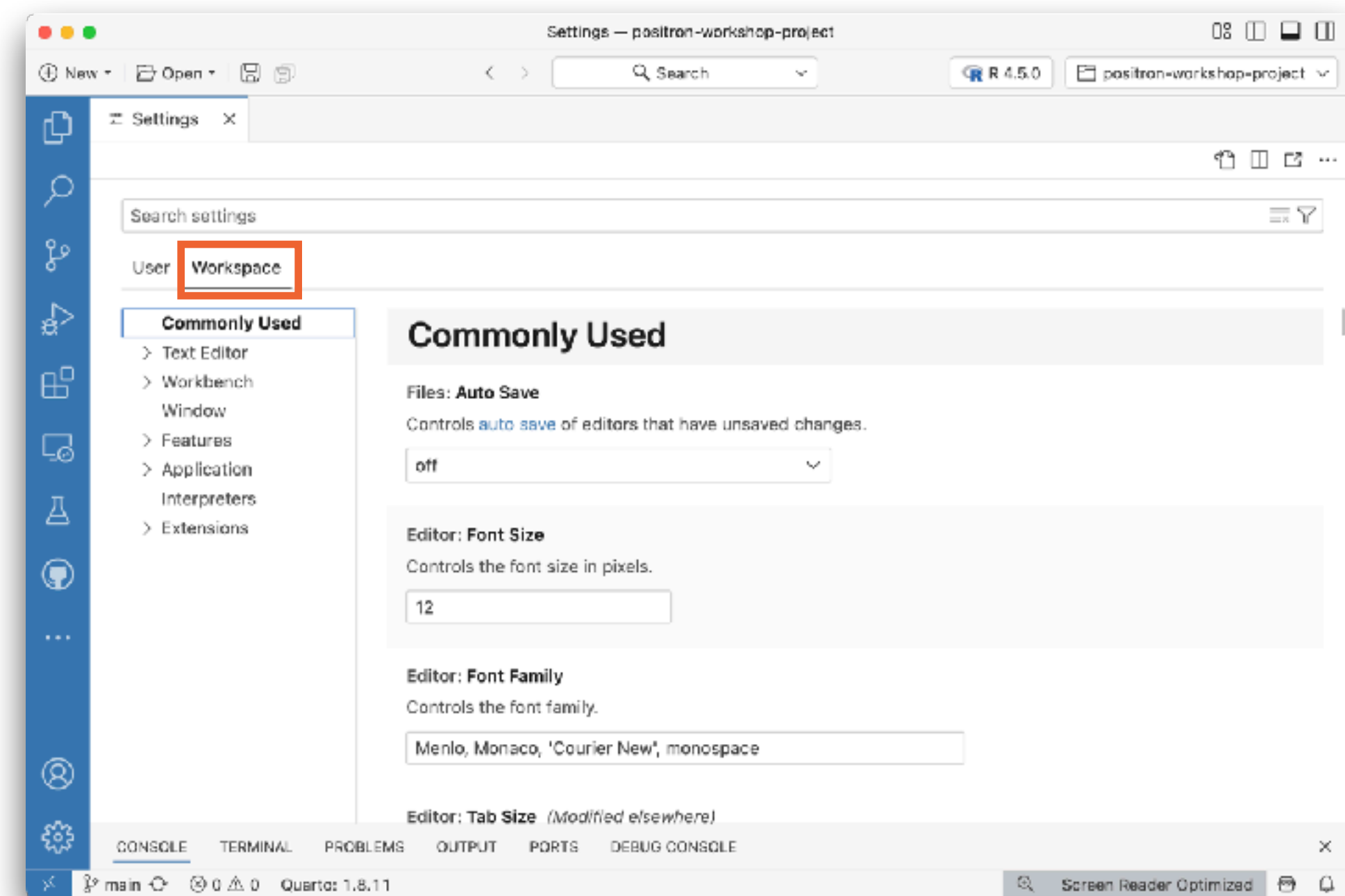
Workspace settings

- Workspace settings are specific to a project and override user settings.
- Positron (like VS Code) stores workspace settings at the root of the project in a `.vscode` folder, making it easy to share settings with others in a version-controlled (e.g., Git) project.

Updating workspace settings

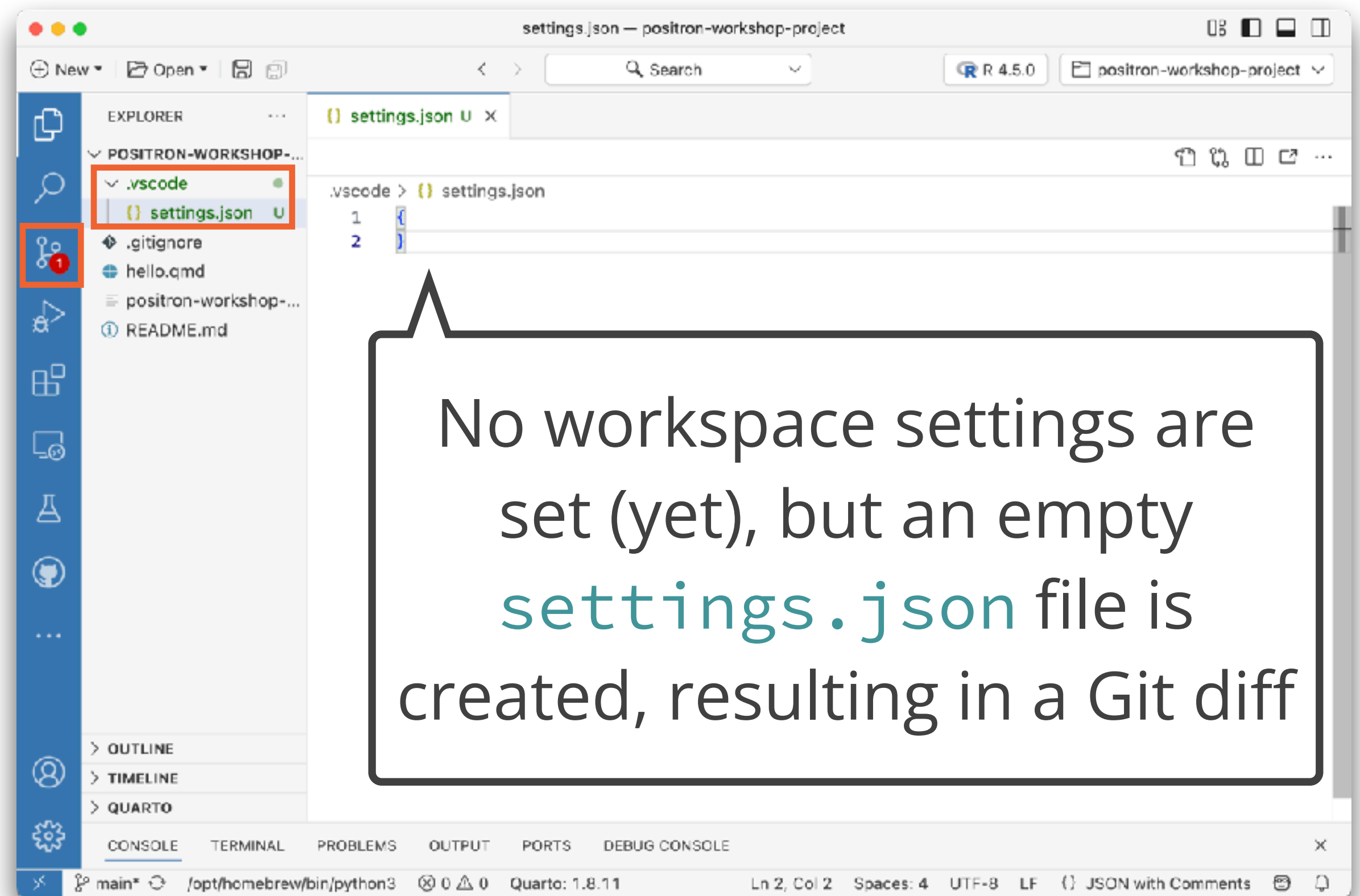
To update using the UI:

- In the Command Palette:
Preferences: Open Workspace Settings
- In the top menu:
Positron > Settings... > Settings



To edit the JSON file:

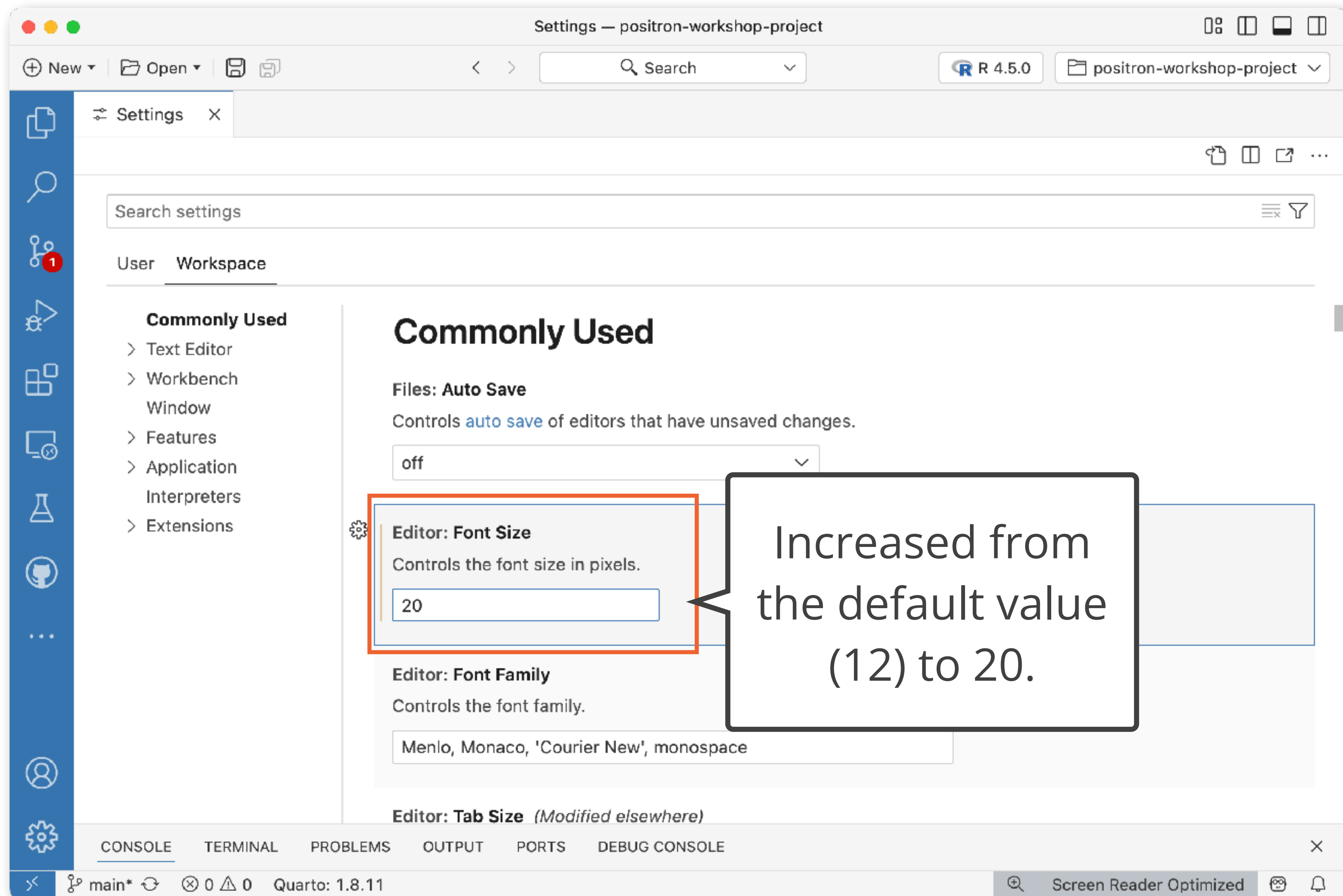
In the Command Palette:
Preferences: Open Workspace Settings (JSON)

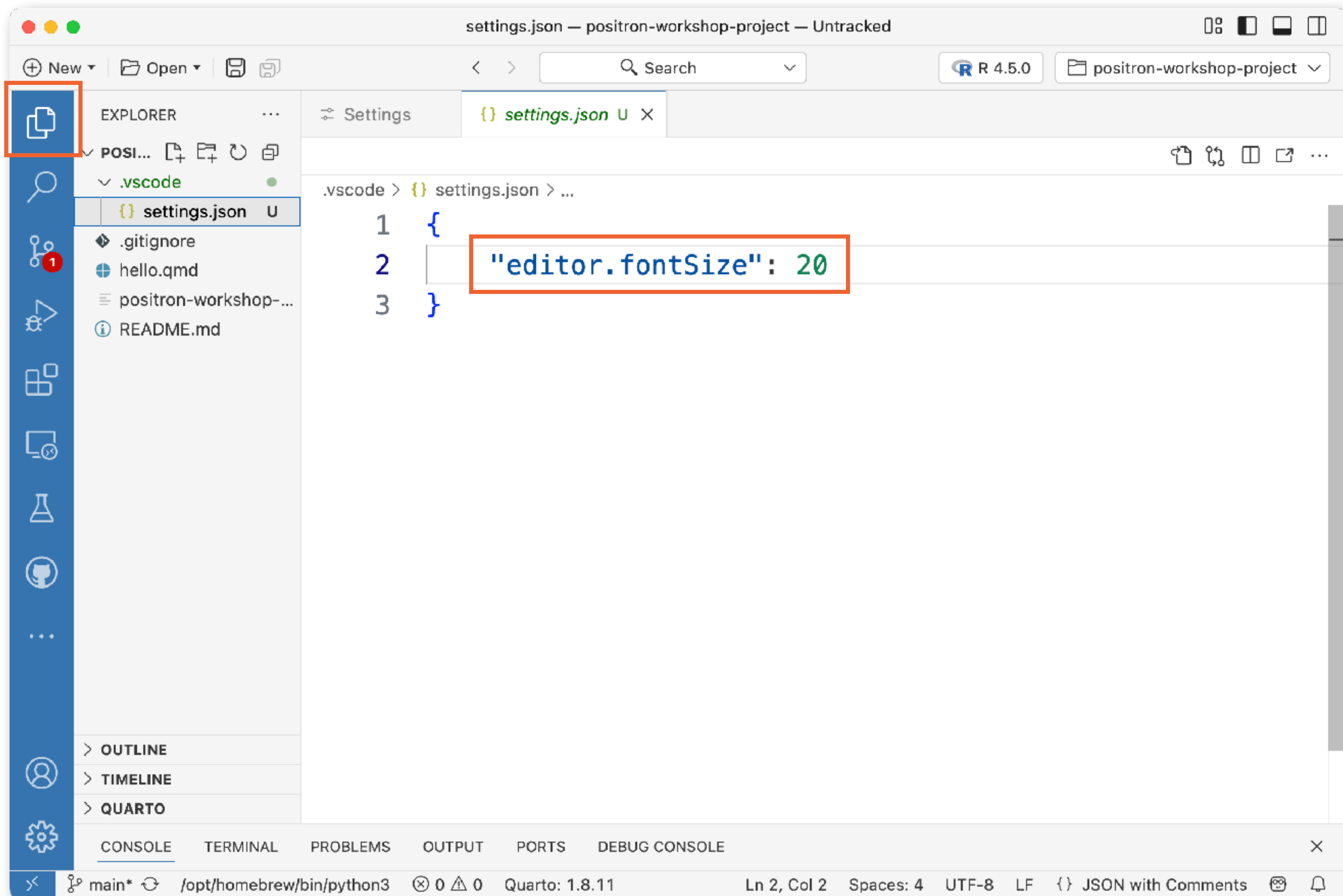


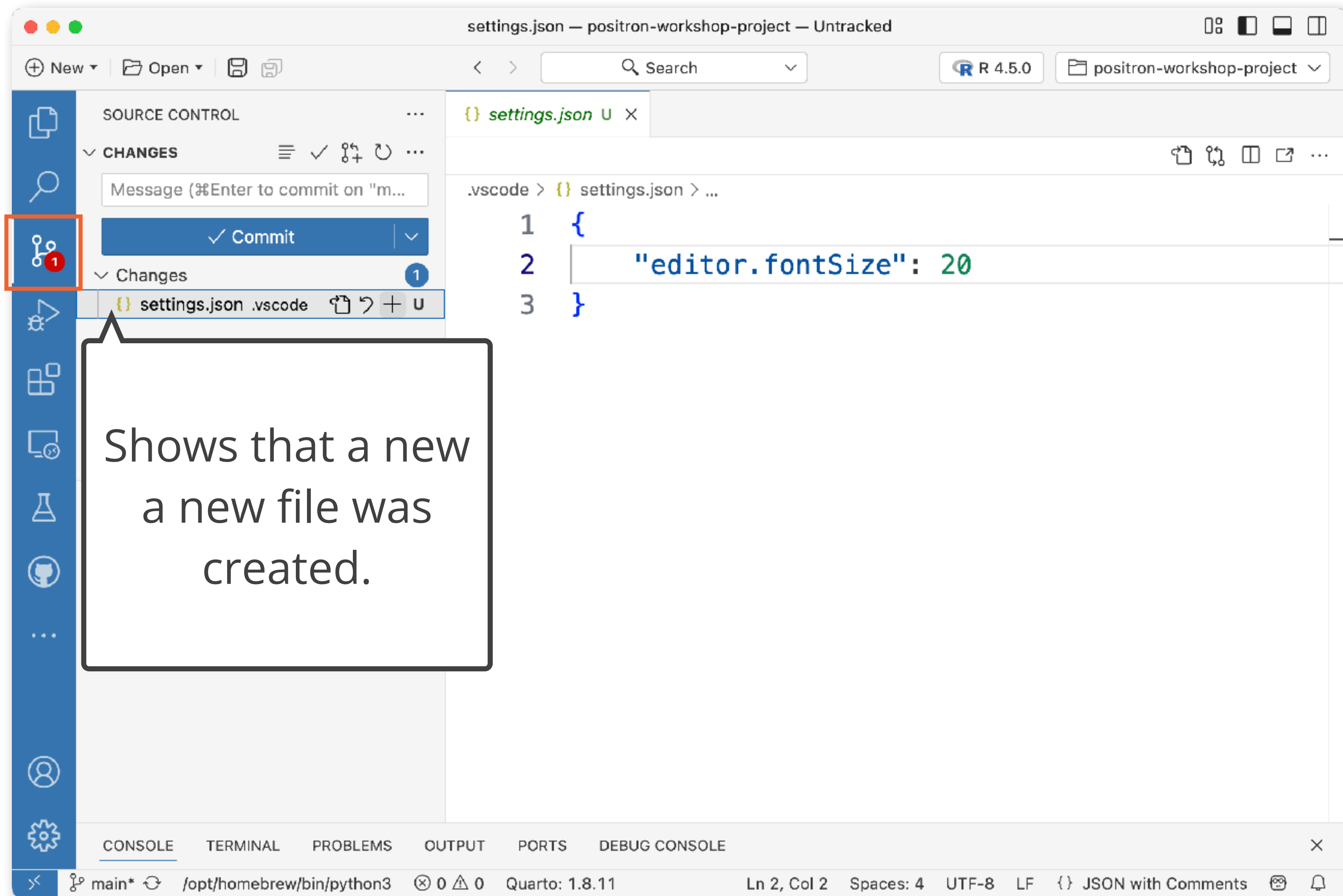
Your turn

Customize workspace settings

- Update your workspace settings in the positron-workshop-project using the Settings editor interface.
 - You can update whatever setting you like.
 - If you need inspiration, change the font size in the editor.
- Open the `settings.json` file in the `.vscode` folder of your project to confirm the change you made.
- Click on the Source Control icon in the Activity bar and view the diff created by updating your settings. Contrast this to what happened (or didn't happen) in the Source Control when you updated your user settings.



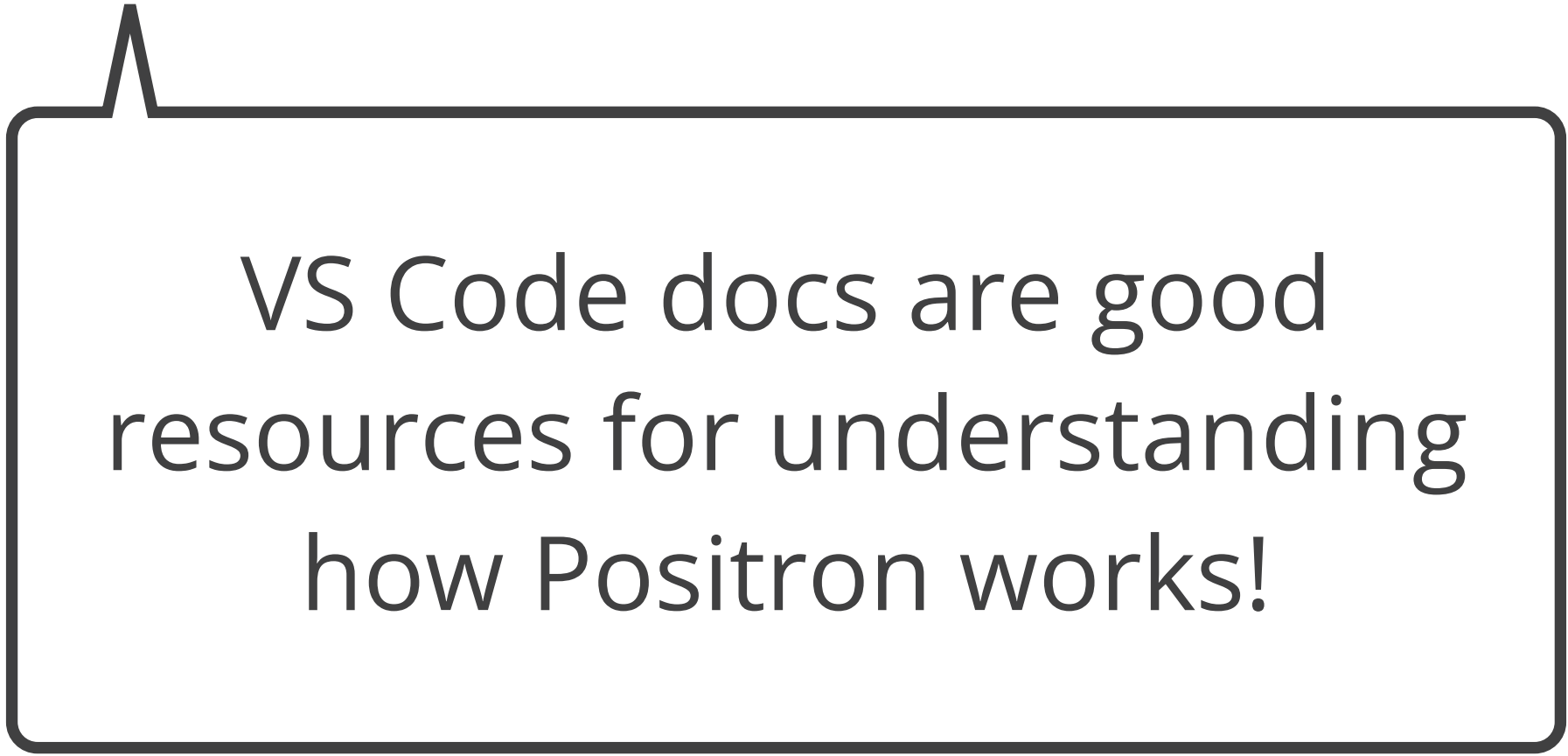




Learn more

VS Code docs: User and workspace settings

code.visualstudio.com/docs/configure/settings



VS Code docs are good
resources for understanding
how Positron works!

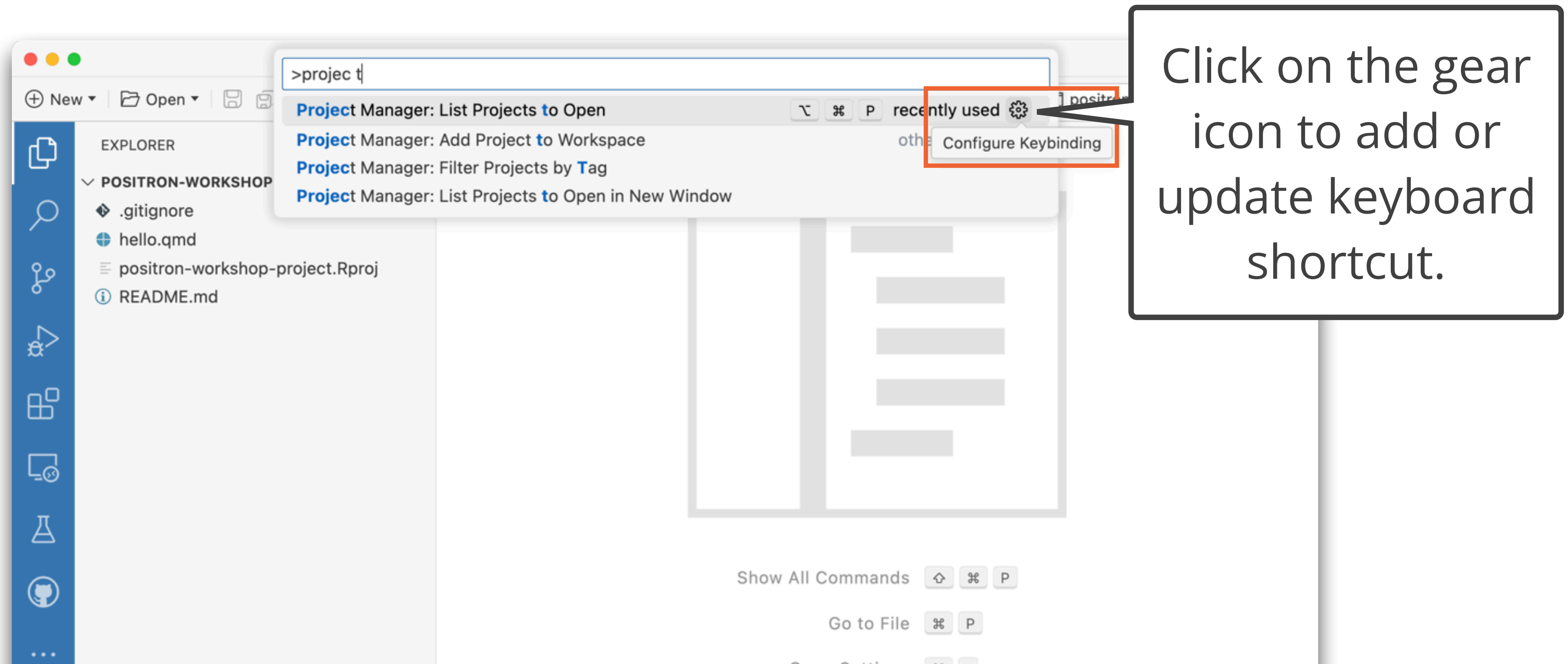
Keyboard shortcuts

Keyboard shortcuts

- Positron lets you perform most tasks directly from the keyboard.
- The Keyboard Shortcuts editor lists all available commands with and without keyboard shortcuts, and enables you to change, remove, or reset their keyboard shortcuts using the available actions.
- You can open the Keyboard Shortcuts editor:
 - From the Command Palette: *Preferences: Open Keyboard Shortcuts*
 - From the top menu: *Code > Settings > Keyboard Shortcuts*

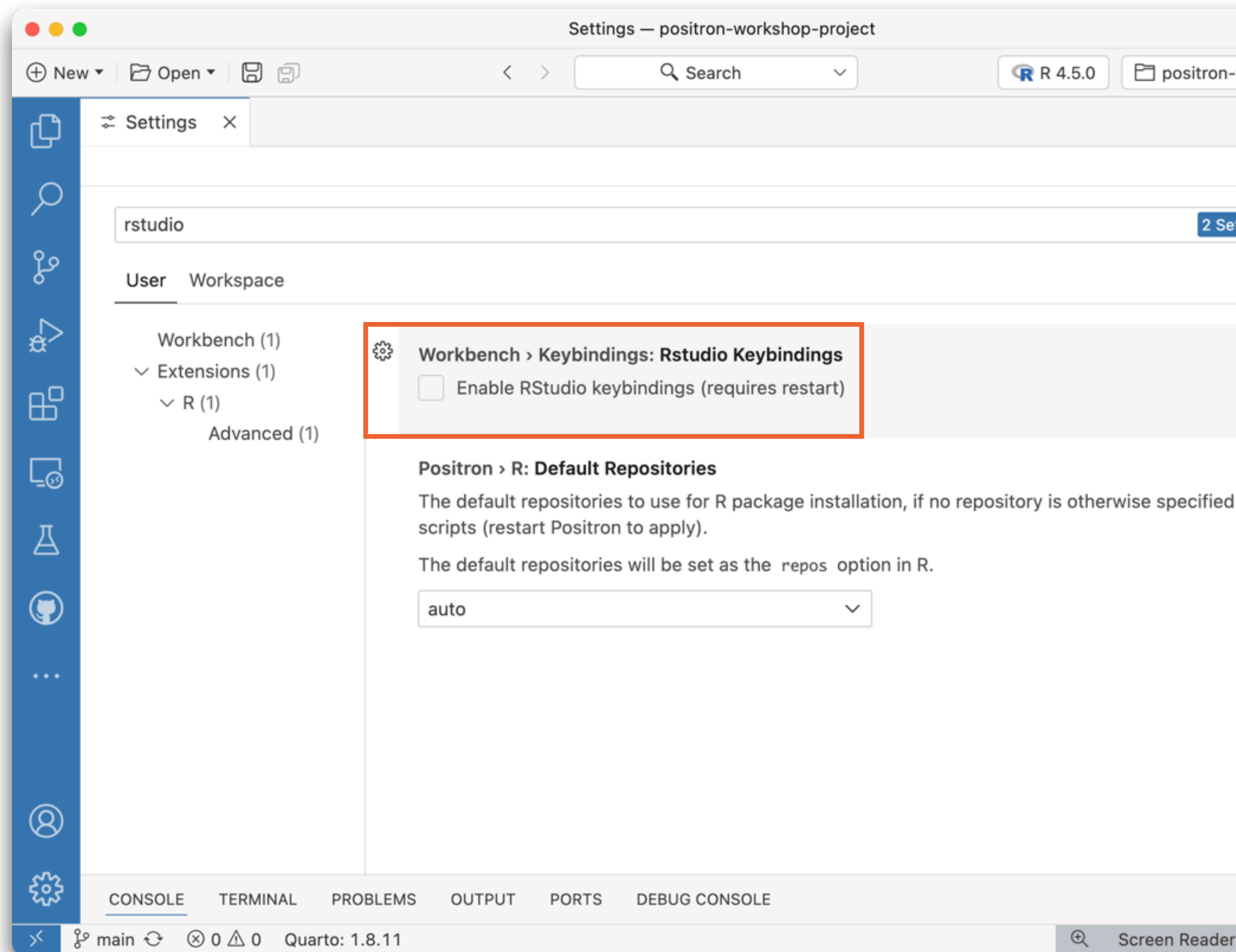
Keyboard shortcuts for Command Palette actions

You can add a keyboard shortcut or customize the existing keyboard shortcut for any action you can take from the Command Palette.



Your RStudio keybindings in Positron

- You can enable RStudio keybindings in Positron with the Enable RStudio keybindings option.
- This might overwrite some of Positron's keybindings, so you might then wish to update / remove some newly enabled RStudio keybindings.



Learn more

VS Code docs: Keyboard shortcuts for Visual Studio Code

code.visualstudio.com/docs/config/keybindings

Layout

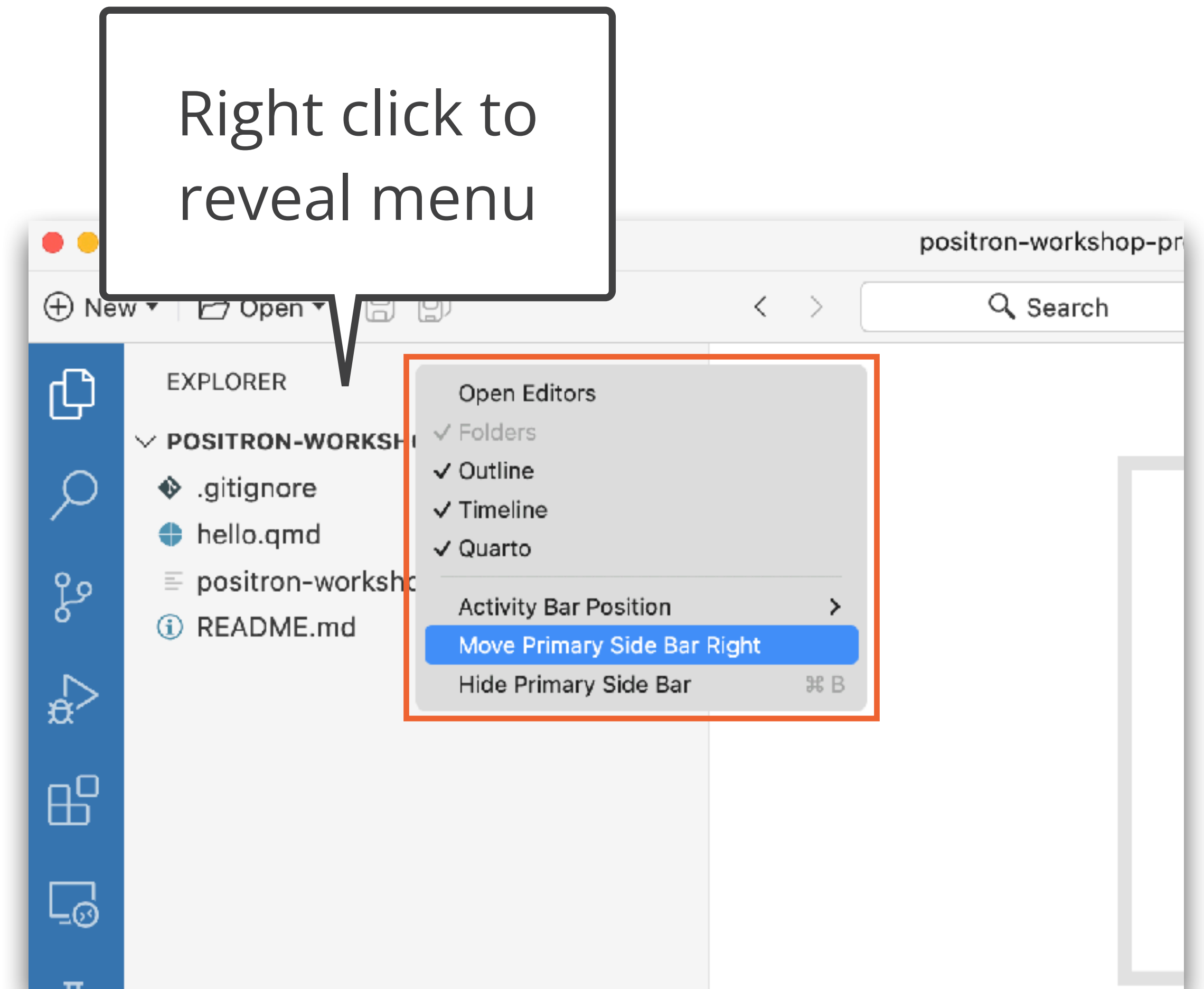
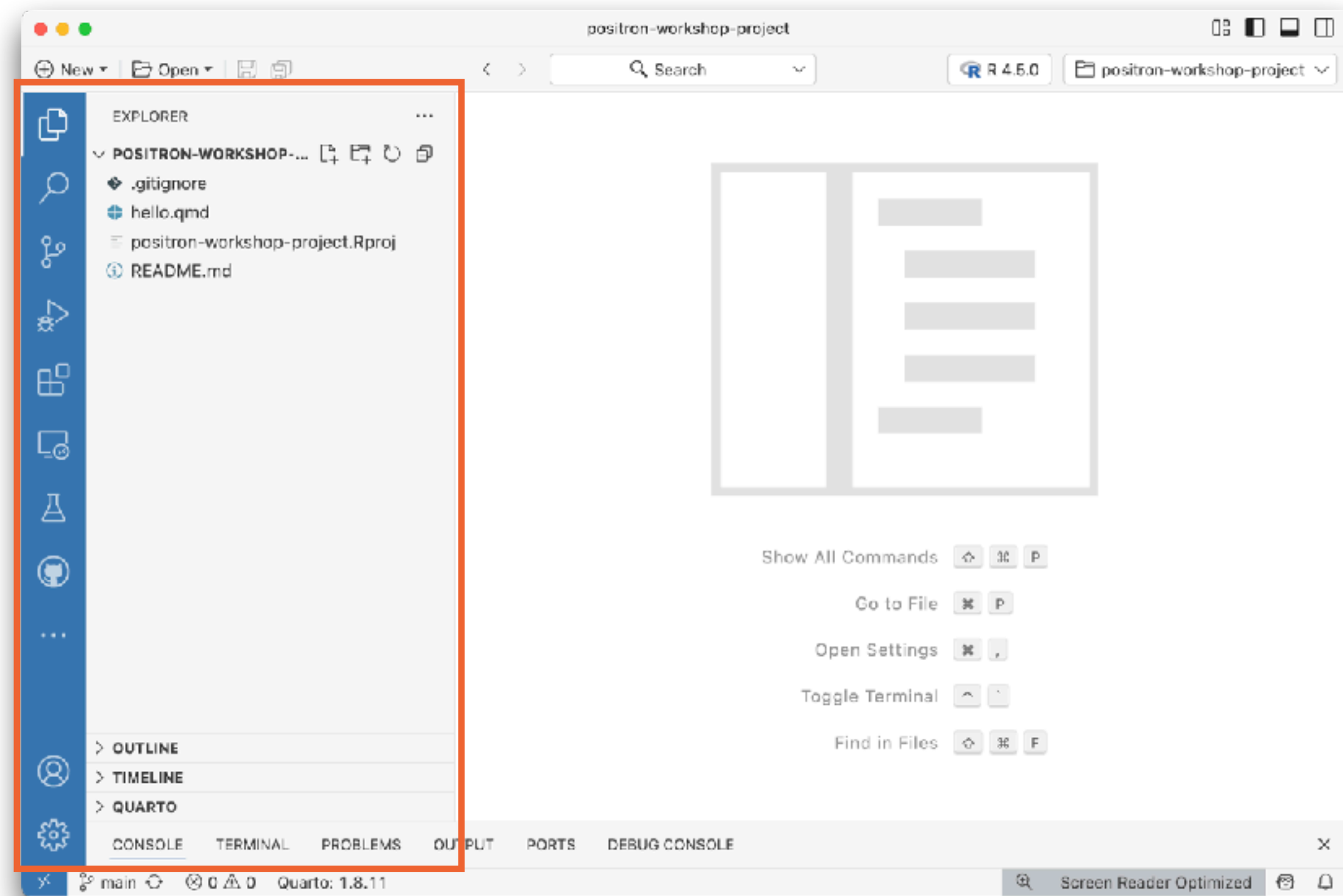
Primary side bar

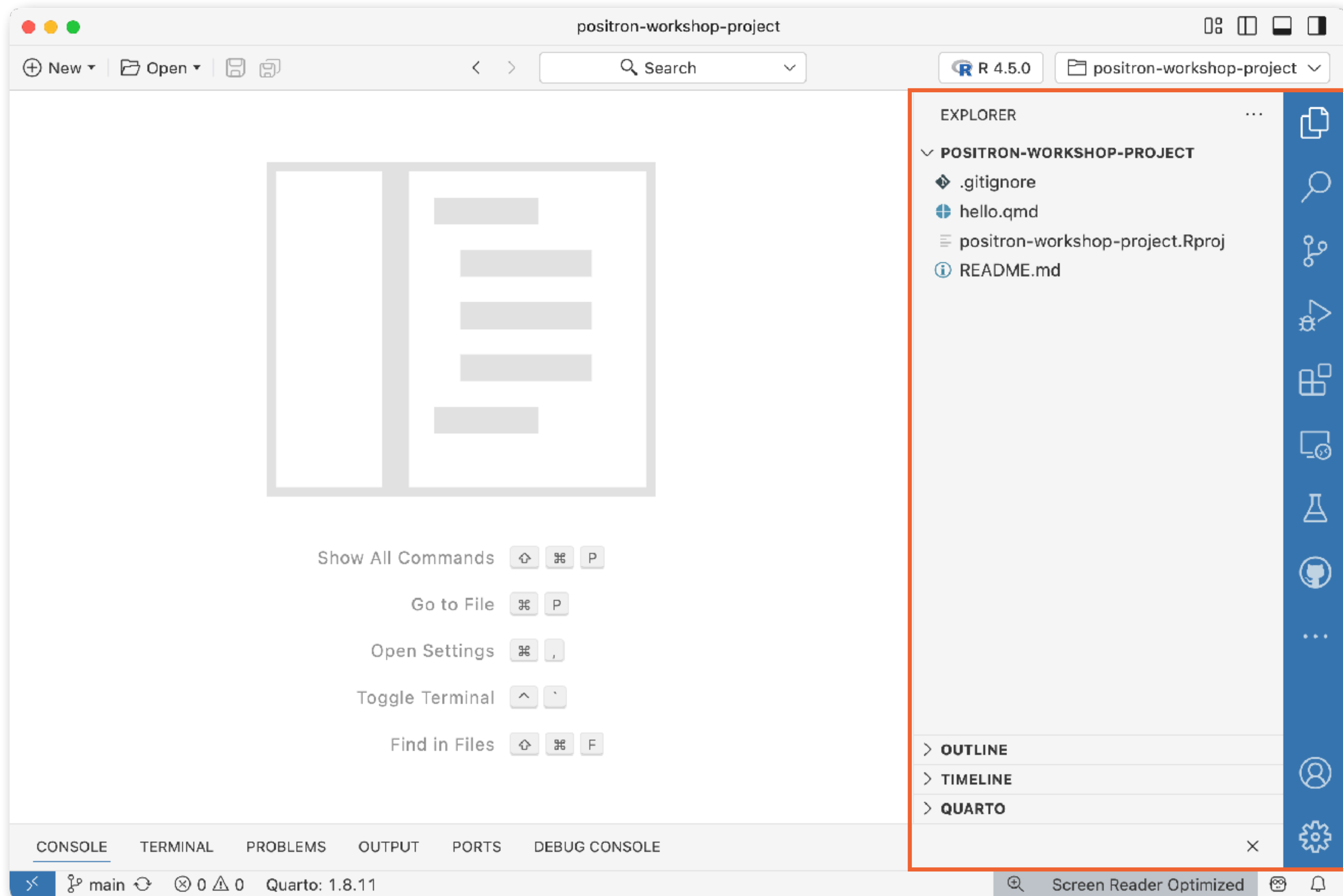
- By default, the Primary Side Bar is located on the left and shows views such as the Explorer, Search, and Source Control views.
- Contents of the Primary Side Bar is controlled by what is selected in the Activity Bar.

Demo

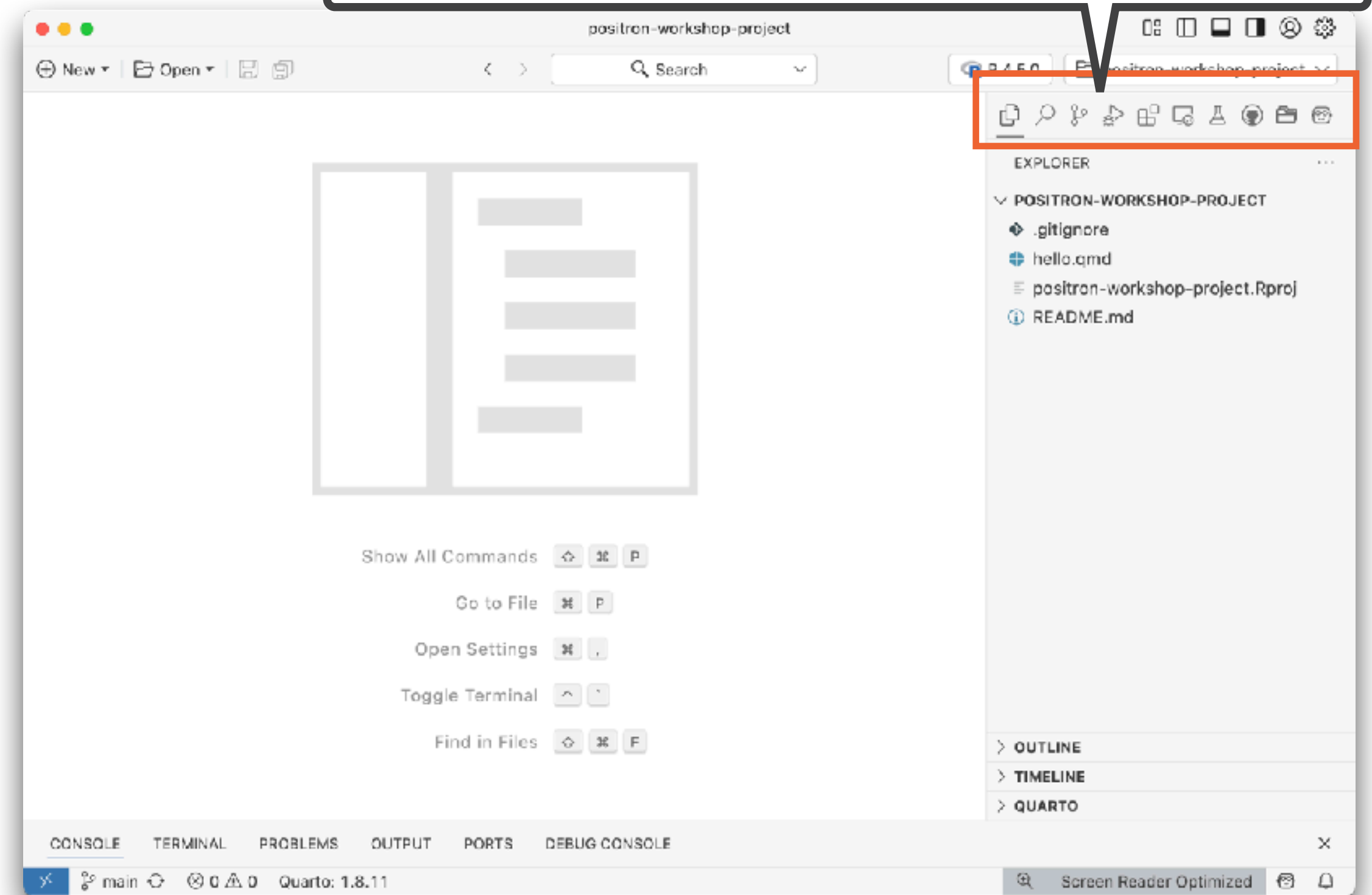
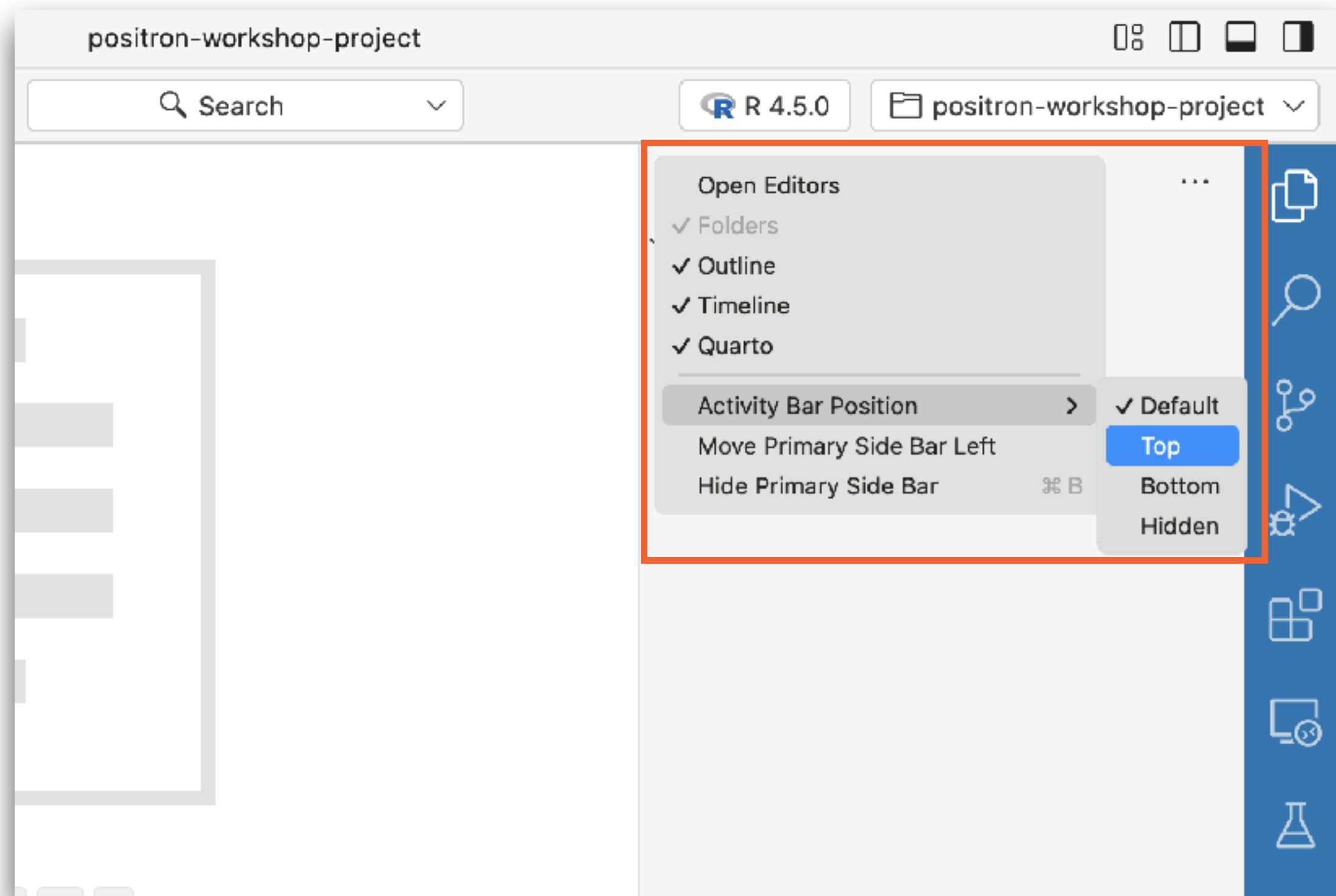
Customizing layout

- Move the primary side bar to the right.
- Move the activity bar to the top.



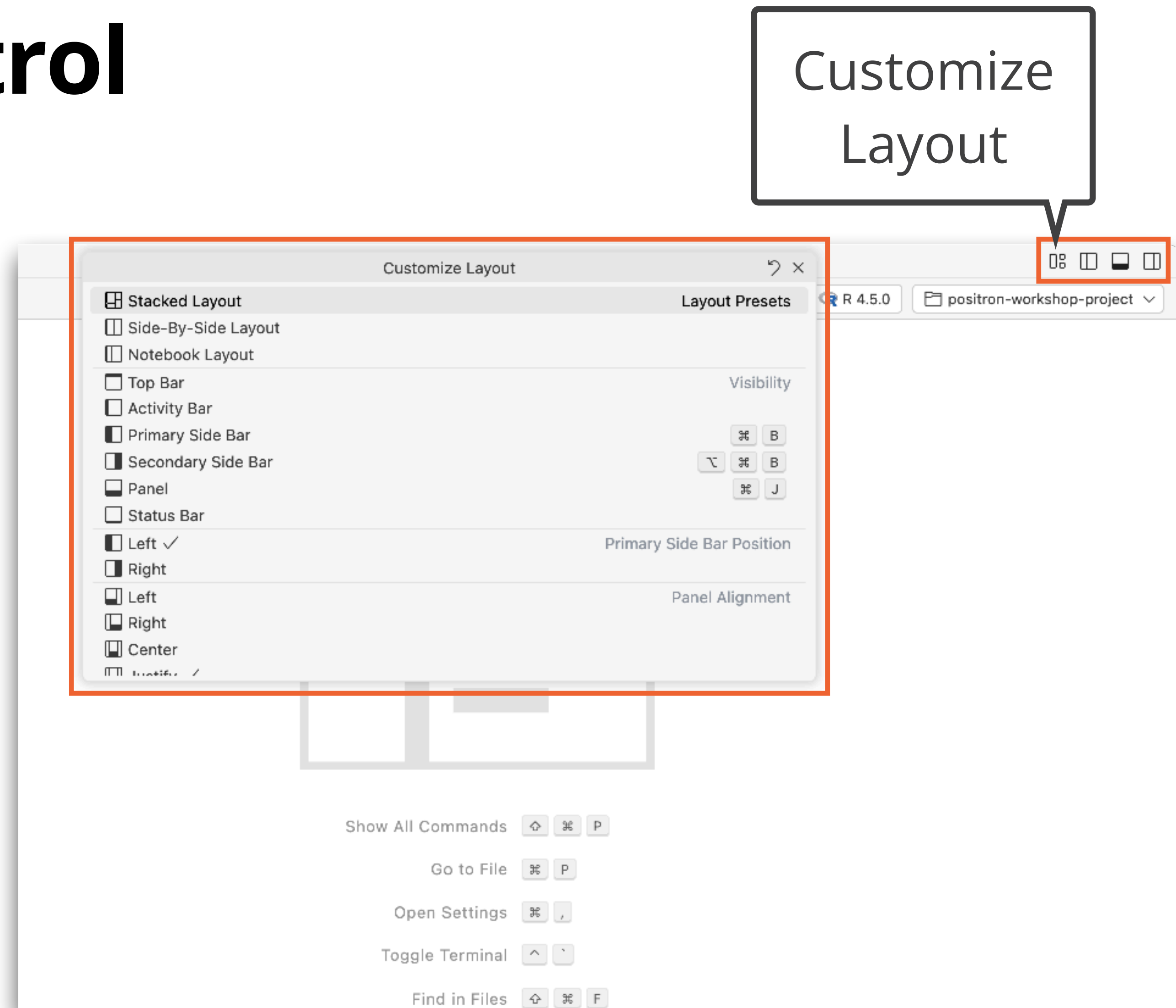


Activity Bar on top,
and not blue anymore.



Customize layout control

- The title bar has buttons to toggle the visibility of UI elements.
- The rightmost button brings up the Customize Layout dropdown that gives you finer control over the visibility and layout of various UI elements.





Teaching tip

Side-by-side layout

Quarto document / editor — Full height

Console — Prompt visible from the back of the classroom

Variables and data

Plots — that can be popped out into the editor / new window for inspection

The screenshot displays the Positron IDE interface with a Quarto document open. The document is titled 'hello.qmd' and shows a preview of a Quarto document. The preview includes a title 'Hello, Quarto', a subtitle 'Dimensions for penguins at Palmer Station LTER', and a plot titled 'Flipper and bill length'. The plot shows the relationship between flipper length (mm) on the x-axis and bill length (mm) on the y-axis, with points colored by penguin species: Adelle (orange), Chinstrap (purple), and Gentoo (teal). The console on the right shows the R code used to create the plot, including the `scale_color_manual` function. The variables panel shows the data frame `penguins` with 344 rows and 8 columns. The plot panel shows the plot titled 'Flipper and bill length' with dimensions for penguins at Palmer Station LTER.

```
title: "Hello, Quarto"
format: html
editor: visual

{r}
1 #| label: load-packages
2 #| include: false
3 library(tidyverse)
4 library(palmerpenguins)

Meet Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

Meet the penguins

The penguins data from the palmerpenguins package contains size measurements for {r}
nrow(penguins)
penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.

The plot below shows the relationship between flipper and bill lengths of these penguins.

{r}
1 #| label: plot-penguins
2 #| warning: false
3 #| echo: false
```

```
+ scale_color_manual(values = c("darkorange", "purple", "cyan4")) +
+ labs(
+   title = "Flipper and bill length",
+   subtitle = "Dimensions for penguins at Palmer Station LTER",
+   x = "Flipper length (mm)", y = "Bill length (mm)",
+   color = "Penguin species", shape = "Penguin species"
+ ) +
+ theme_minimal()
Warning message:
Removed 2 rows containing missing values or values outside the scale range
('geom_point()').
>
```

VARIABLES

R 4.5.1

DATA

penguins [344 rows x 8 columns] <tbl_df>

	species	island	bill length mm
>	Adelle	Torgersen	39.1
>	Adelle	Torgersen	39.5
>	Adelle	Torgersen	40.3
>	Adelle	Torgersen	NA
>	Adelle	Torgersen	36.7
>	Adelle	Torgersen	39.3
>	Adelle	Torgersen	38.9
>	Adelle	Torgersen	39.2
>	Adelle	Torgersen	34.1
>	Adelle	Torgersen	42
>	Adelle	Torgersen	37.8
>	Adelle	Torgersen	37.8
>	Adelle	Torgersen	41.1
>	Adelle	Torgersen	38.6
>	Adelle	Torgersen	34.6
>	Adelle	Torgersen	36.6
>	Adelle	Torgersen	38.7
>	Adelle	Torgersen	42.5
>	Adelle	Torgersen	34.4
>	Adelle	Torgersen	...

Plots

Flipper and bill length
Dimensions for penguins at Palmer Station LTER

Bill length (mm)

Flipper length (mm)

Penguin species

- Adelle
- Chinstrap
- Gentoo

Learn more

VS Code docs: Custom Layout

code.visualstudio.com/docs/config/custom-layout

Extensions

Extensions

- Extensions let you add languages, debuggers, tools, themes, etc. to Positron to support your development workflow (just like in VS Code).
- Your Positron and VS Code extensions are separate and won't interfere with each other.
- Unlike in VS Code, you install Positron extensions from Open VSX.

Your turn

Install extension

- Click on the Extensions icon from the Activity Bar in Positron.
- If you haven't yet done so, search for and install the Shiny, or Project Manager extension.
- If you have already installed it, or if you would like to install one more extension, find and install the Solarized & Selenized extension.
- Use the Command Palette to set the theme of your Positron window to a Solarized or Selenized theme of your choice. **Hint:** Start typing *theme* in the Command Palete.

Extensions

Welcome — positron-workshop-project

⊕ New ▾ | 📁 Open ▾ | 📄 | 📄 | 🔍 Search ▾ | R 4.5.0 | positron-workshop-project ▾

EXTENSIONS 🔄 ...

Search Extensions in Marke... ≡ 🔍

✓ **INSTALLED** 4

-  **Air - R Language Su...** ⌚ 17ms
R formatter and language ser...
Posit Software, PBC ⚙️
-  **GitHub Pull Requests** ⌚ 33ms
Pull Request and Issue Provi...
GitHub ⚙️
-  **Project Manager** ⌚ 5ms
Easily switch between projects
alefragnani ⚙️
-  **Quarto** ⌚ 128ms
Extension for the Quarto scie...
quarto ⚙️

✓ **RECOMMENDED** 0

Welcome ×

 **Positron**
an IDE for data science

Start

-  New Notebook
-  New File
-  

Open

-  Open...
-  New Folder...
-  New Folder from Git...
-  Connect to...

Recent

- positron-workshop ~/Deskto...

SESSION **CONNECTIONS** ... ×

R 4.5.0 ▾

No variables have been created.

CONSOLE **TERMINAL** **PROBLEMS** **OUTPUT** **PORTS** **DEBUG CONSOLE** ×

✓ **CONSOLE** 🔄 🗑️ 🔍 📄 ≡

~/Desktop/posit/positron-workshop-project

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>

PLOTS 🔄 🗑️ 🔍 📄 ≡

main 🔄 ⊗ 0 ⚠️ 0 Quarto: 1.8.11

Screen Reader Optimized 🗣️ 🔔

Welcome — positron-workshop-project

⊕ New ▾ | 📁 Open ▾ | 📄 📄 | < > | 🔍 Search ▾ | R 4.5.0 | positron-workshop-project ▾

EXTENSIONS: MARKETPLACE ↻ ...

Solarized 🔍

 **Solarized & Selen...** 1K ⭐ 5
A Solarized and Selenized co...
santoso-wijaya [Install](#)

 **Solarized Sunset Th...** 1K
A flat theme based on Solari...
jansim [Install](#)

 **Better Solarized** 12K
A Better Solarized theme for ...
Ed Heltzel [Install](#)

 **Solarized Sharp** 3K
muhammad-sammy [Install](#)

 **Solarized High Contr...** 96
Sepia-toned, high-contrast li...
tiny [Install](#)

Positron
an IDE for data science

Start

 New Notebook

 New File

Open

 Open...

 New Folder...

 New Folder from Git...

 Connect to...

Recent

positron-workshop ~/Desкто...

SESSION CONNECTIONS ...

R 4.5.0 ▾ filter

No variables have been created.

CONSOLE TERMINAL PROBLEMS OUTPUT PORTS DEBUG CONSOLE

CONSOLE

~/Desktop/posit/positron-workshop-project

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Type 'q()' to quit R.

>

main 0 0 Quarto: 1.8.11

Screen Reader Optimized

New

Open

EXTENSIONS: MARKETPLACE

Solarized

Solarized & Selenized

A Solarized and Selenized color themes for VS Code

santoso-wijaya

Solarized Sunset Theme

A flat theme based on Solarized

jansim

1K

Install

Better Solarized

A Better Solarized theme for VS Code

Ed Heltzel

12K

Install

Solarized Sharp

muhammad-sammy

3K

Install

Solarized High Contrast

Sepia-toned, high-contrast light theme

tiny

96

Install

>theme

Preferences: Color Theme

Developer: Generate Color Theme From Current Settings

Preferences: Browse Color Themes in Marketplace

Preferences: File Icon Theme

Preferences: Product Icon Theme

Preferences: Toggle between Light/Dark Themes

other commands

Solarized and Selenized for VS Code

santoso-wijaya

1,269

★★★★★ (1)

A Solarized and Selenized color themes for VS Code

Set Color Theme

Disable

Uninstall

✓ A

DETAILS

FEATURES

CHANGELOG

Solarized and Selenized for VS Code

Solarized is a sixteen color palette (eight colors) and Selenized is a sixteen color palette (eight colors) for VS Code.

Marketplace

Identifier santoso-wijaya.helios-selene

Version 0.3.18

Publish... 2024-07-08, 22:02:59

CONSOLE

TERMINAL

PROBLEMS

OUTPUT

PORTS

DEBUG CONSOLE

CONSOLE

~/Desktop/posit/positron-workshop-project

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Type 'q()' to quit R.
>

PLOTS

R 4.5.0

positron-workshop-project

SESSION

CONNECTIONS

filter

No variables have been created.

main

Quarto: 1.8.11

Screen Reader Optimized

New

Open

EXTENSIONS: MARKETPLACE

Solarized



Solarized & Selenized

A Solarized and Selenized color theme for VS Code

santoso-wijaya

Install



Solarized Sunset Theme

A flat theme based on Solarized

jansim

Install



Better Solarized

A Better Solarized theme for VS Code

Ed Heltzel

Install



Solarized Sharp

A Solarized theme with sharp colors

muhammad-sammy

Install



Solarized High Contrast

Sepia-toned, high-contrast light theme

tiny

Install

Select Color Theme (detect system color mode disabled)

+ Browse Additional Color Themes...

Positron Light Default Positron Light

Selenized Light Selene Selenized Light

Solarized Light Helios Solarized Light

light themes

Positron Dark Default Positron Dark

Selenized Dark Selene Selenized Dark

Solarized Dark Helios Solarized Dark

Tomorrow Night Blue

dark themes

Light High Contrast Default High Contrast Light

Dark High Contrast Default High Contrast

high contrast themes

DETAILS

FEATURES

CHANGELOG

Solarized and Selenized for VS Code

Solarized is a sixteen color palette (eight colors) and Selenized is a sixteen color palette (eight colors) with a different color palette.

Marketplace

Identifier

santoso-wijaya.helios-selene

Version

0.3.18

Publish...

2024-07-08, 22:02:59

CONSOLE

TERMINAL

PROBLEMS

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DEBUG CONSOLE

CONSOLE

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>

PLOTS

R 4.5.0

positron-workshop-project

SESSION CONNECTIONS

R 4.5.0 filter

No variables have been created.

main

Quarto: 1.8.11

Screen Reader Optimized

Extension: Solarized & Selenized — positron-workshop-project

Search

R 4.5.0 positron-workshop-project

EXTENSIONS: MARKETPLACE

Solarized

Solarized & Selenized
A Solarized and Selenized co...
santoso-wijaya

Solarized Sunset Th... 1K
A flat theme based on Solari...
jansim **Install**

Better Solarized 12K
A Better Solarized theme for ...
Ed Heltzel **Install**

Solarized Sharp 3K
muhammad-sammy **Install**

Solarized High Contr... 96
Sepia-toned, high-contrast li...
tiny **Install**

Solarized & Selenized
santoso-wijaya | 1,269 | ★★★★★ (1)
A Solarized and Selenized color themes com...
Set Color Theme **Disable** **Uninstall** ☒ A

DETAILS FEATURES CHANGELOG

Solarized and Selenized for VS Code

Marketplace

Identifier santoso-wijaya.helios-selene

Version 0.3.18

Publish... 2024-07-08, 22:02:59

CONSOLE

~/Desktop/posit/positron-workshop-project

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'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>

main 0 0 Quarto: 1.8.11

Screen Reader Optimized

Solarized
Dark
theme

Built-in extensions

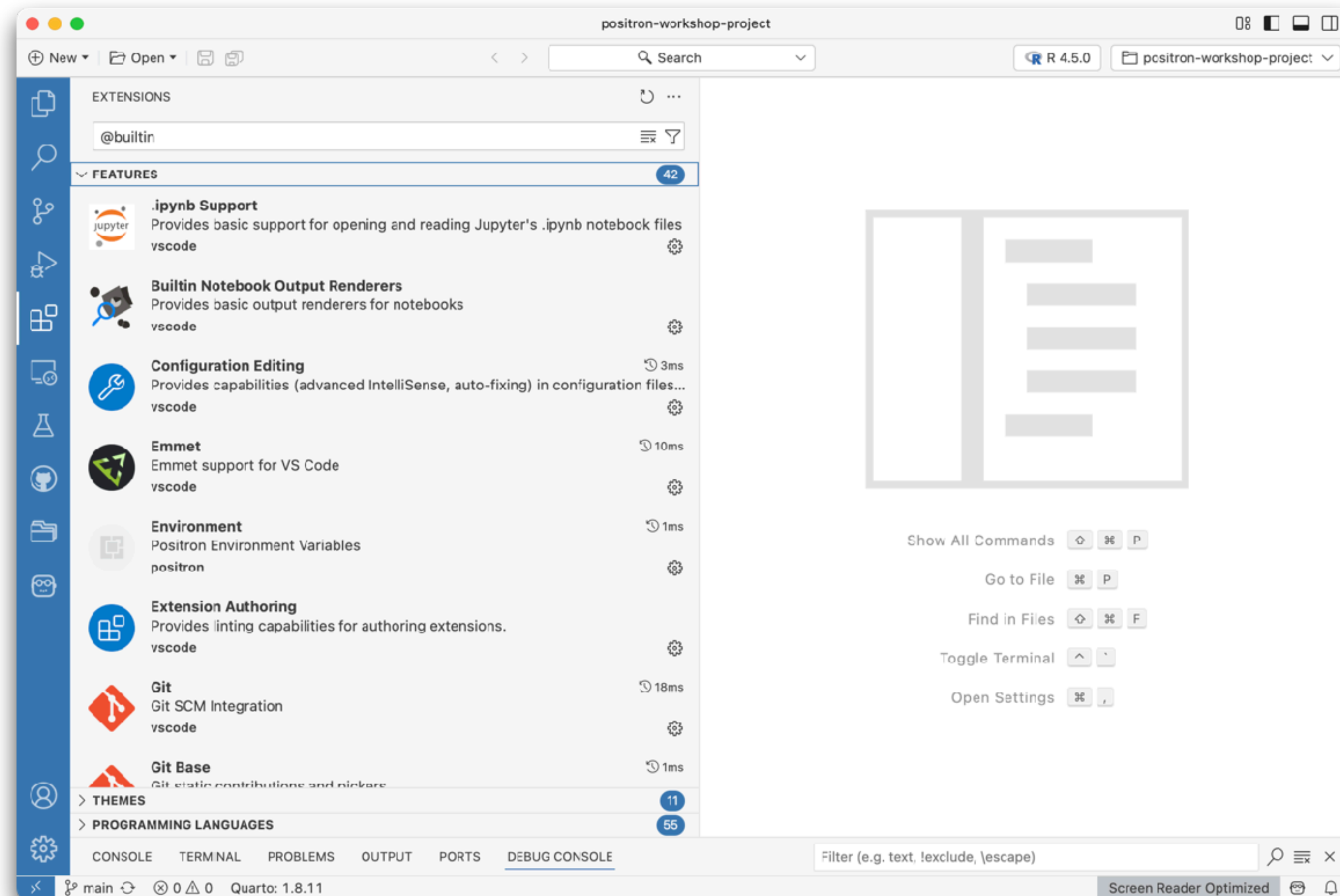
- Built-in extensions (e.g., R, Positron Python, and Markdown Language Basics) are installed with Positron.
- They can only be updated by updating Positron.
- They can be seen in the extensions view with [@builtin](#).

Demo

Built-in extensions

- Click on the Extensions icon in the Activity Bar.
- Type `@builtin` in the Extension search bar.
- Review the built-in extensions.

Built-in extensions



Bundled extensions

- Bundled extensions (e.g., Quarto, Air, GitHub Pull Requests) are installed with Positron.
- They can be updated independently of updating Positron.

Other extensions

- Any other extensions need to be installed from Open VSX.
- Popular extensions you might consider installing are Project Manager, Shiny, Rainbow CSV, indent-rainbow, etc.
- You can also install Extension Packs; a collection of packs curated by users for specific tastes, tasks, or languages.

5_m 00_s

Your turn

Install the ugliest theme extension

- Find the ugliest theme extension you can.
- Install the extension and apply the theme.
- **Note:** You might get a warning about whether you trust the publisher of a theme you choose. Read the warning carefully. The decision is yours!

New

Open

Search

R 4.5.0

positron-workshop-project

EXTENSIONS: MARKETPLACE

theme



Solarized & Selenized

A Solarized and Selenized colo...

santoso-wijaya

Dracula Theme Official

The official Dracula Theme: a ...

dracula-theme

Installing

base9-vscode

base9-theme

2K

Install

Wildberries

A dark theme for purple lovers

wildberries-theme

7K 4

Install

Material Theme (But I ...

A Apache-2 licensed fork of M...

t3dotgg

2K

Install

Cobalt2 Theme O...

Official theme by Wes Bos.

wesbos

19K 5

Install

XD Theme

A theme because my settings ...

jeff-hykin

4K

Install

OpenSumi Default Th...

The default OpenSumi themes

opensumi

14K

Install

MSDevStudio Default...

745

Extension: Dracula Theme Official



Dracula Theme Official

dracula-theme | 158,746 | (2)

The official Dracula Theme: a dark theme for many editors, shells, and more. ...

Do you trust the publisher "dracula-theme"?

The extension [Dracula Theme Official](#) is published by dracula-theme. This is the first extension you're installing from this publisher.

dracula-theme is **not verified**.

Positron has no control over the behavior of third-party extensions, including how they manage your personal data. Proceed only if you trust the publisher.

Learn More

Cancel

Trust Publisher & Install

Marketplace

Identifier

dracula-theme.theme-dracula

Version

2.25.1

Publish...

2020-04-06, 23:07:47

Last Released

2024-09-05, 23:29:53

Categories

Themes

Resources

[Marketplace](#)
[Repository](#)

CONSOLE

TERMINAL

PROBLEMS

OUTPUT

PORTS

DEBUG CONSOLE

Filter (e.g. text, \exclude, \escape)

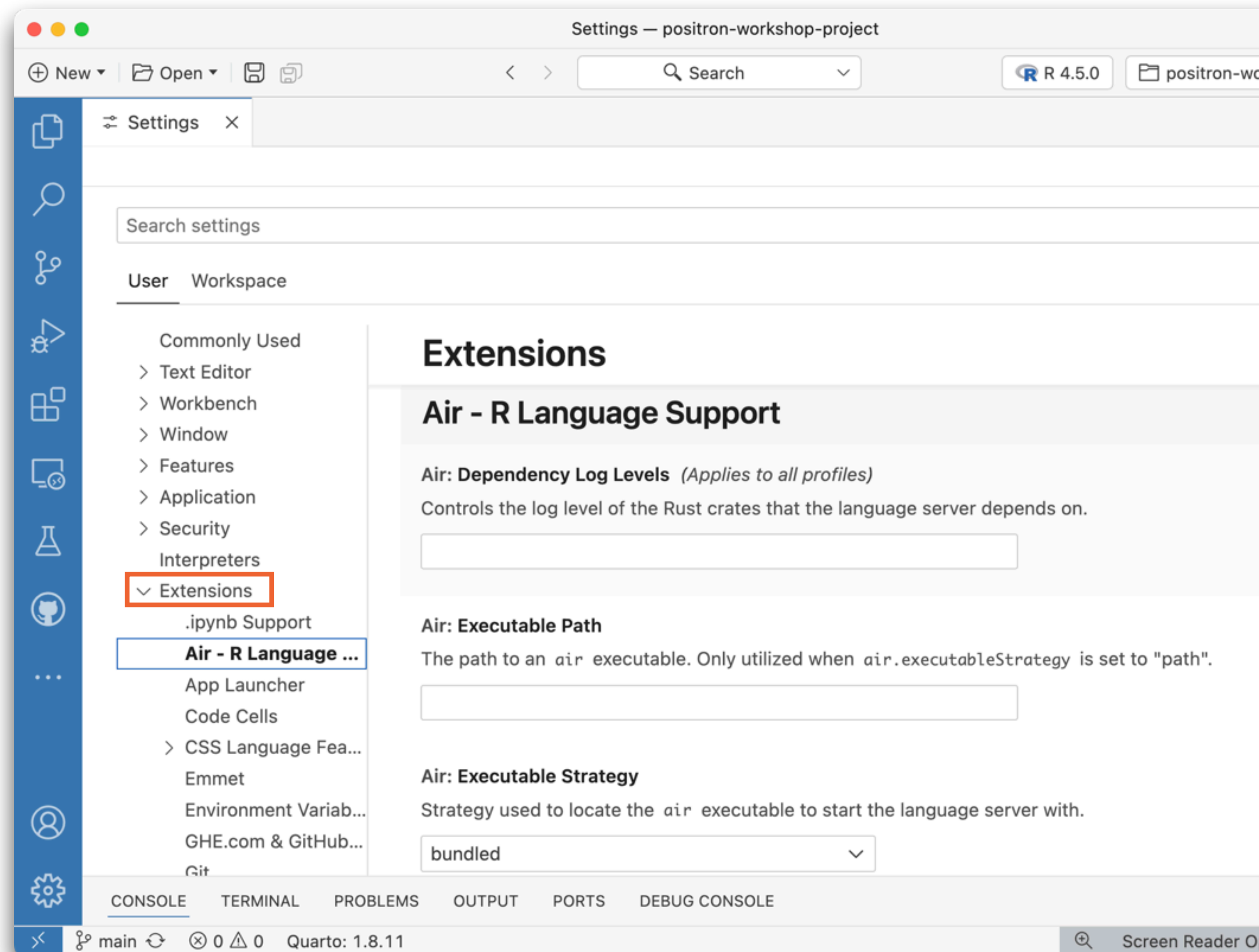
main

Quarto: 1.8.11

Screen Reader Optimized

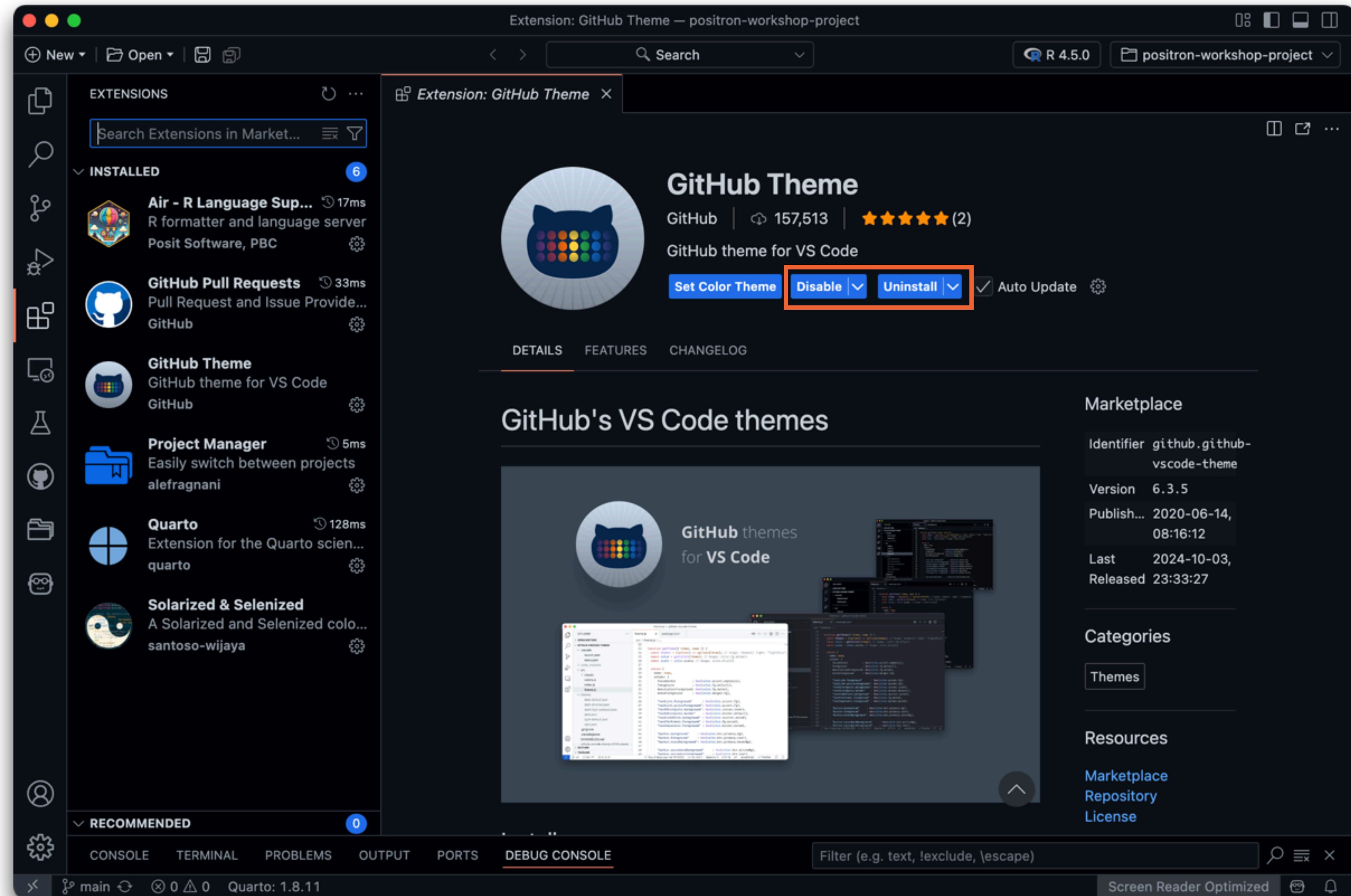
Extension settings

- Installed Positron extensions can also contribute their own settings, which you can review under the Extensions section of the Settings editor.
- These can be set at the user or workspace level as well.



A word of caution

Extensions alter the look and functionality of Positron and can conflict with each other. We recommend being selective in extensions you have enabled at any given point, and disabling or uninstalling the ones you no longer use.



Learn more

Positron docs: Extensions

positron.posit.co/extensions.html